LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fuel, major classification of fuel based on fuel state, turbine engine fuels
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fuel, major classification of fuel based on fuel state, turbine engine fuels
5. Specific Objectives	23.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuel system diagrams; Safety: No handling of fuel samples; Instructor supervision: Monitor diagram interpretation accuracy
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft fuel?
(b) Name two common states of fuel used in aviation.
(c) Why is fuel quality critical for turbine engines?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Liquid vs Solid fuel classification • Jet A and Jet A-1 specifications • Fuel density and freezing points • Additives and corrosion inhibitors • Storage and contamination risks	L & Ds	PPT, IB	15 min
(b)	• Turbine engine fuel requirements • Ignition quality and flash point • Thermal stability under load • Water separation mechanisms • Filtering and straining systems	L & Ds	PPT, PDF	20 min
(c)	• Fuel system component overview • Tank venting and pressure control • Fuel pumps and flow regulation • Fuel heaters and de-icing • Emergency fuel shutoff valves	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between Jet A and Jet A-1 regarding freezing points.
(b) Describe how a turbine engine fuel system prevents water accumulation.
(c) Identify three key components in a typical aircraft fuel system.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between reciprocating engine fuels and turbine engine fuels to review before Lesson Plan 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Brief description and purpose of inverted flight valve and recuperate.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min